

Project Title: Ecommerce Data Vault using Delta Lake and PySpark

Reviewer: Balu Lakshmi Sudha

Review Date: 27-07-2025

1. Documentation

This project demonstrates building an **Ecommerce Data Vault architecture** using **Apache Spark** and **Delta Lake** with the following layers:

- **Raw Layer** – CSV files loaded into Delta
- **Vault Layer** – Hub, Link, and Satellite tables
- **PIT Layer** – For fast querying
- **Delta Lake Features** – Time travel, vacuum, optimize, SCD type 2 support

Data Vault Components

Hubs

hub_customers: Created on customer_id

hub_products: Created on product_sku

Link

link_transaction: Combines transaction_id, customer_id, and product_sku

Satellites

sat_customer: Descriptive customer attributes

sat_product: Product details

sat_transaction: Transaction facts

Peer Review & Documentation Summary

Step 1: Initial Model Review

Model Reviewed:

Data Vault 2.0 architecture with raw → hub/link/satellite layering using Delta Lake.

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Feedback:

The core structure of the model (hubs, links, and satellites) aligns with Data Vault best practices.

Suggested enhancements for:

Consistency of hash key generation across all entities.

Addition of end_time, is_current, and record_hash to satellite tables to support SCD Type 2 logic.

Creation of PIT (Point-In-Time) tables to simplify joins in analytics layer.

Step 2: Model Refinement Based on Feedback

Refinements Implemented:

Satellite tables updated with:

- end_time for versioning close
- is_current flag for current row filtering
- record_hash column to detect attribute-level changes
- Hash keys (*_HK) standardized using SHA-256 with concat_ws.
- Performed DESCRIBE HISTORY, RESTORE, and VACUUM operations to ensure Delta Lake capabilities are fully integrated

Step 3: Final Validation

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Validated Elements:

Hubs and Links are properly defined using unique business keys.

Satellite SCD logic supports history tracking with correct hash comparisons.

Raw → Vault pipeline is deterministic and auditable.

Delta-specific optimizations (OPTIMIZE ZORDER, VACUUM, RESTORE) verified.

